CS 473 (Spring 2025)

Yiming Qin (yimingq3@illinois.edu)

Problem 6.3

Solution:

Idea: According to Berge's theorem, if there is no augmenting path in the current bipartite graph, then the current matching has already reached the maximum matching size.

Algorithm 1

Require: Bipartite graph $G = (X \cup Y, E)$, a maximum matching M in G, and a vertex $v \in V(G)$ to delete.

Ensure: A maximum matching M' in the new graph $G' = G \setminus \{v\}$.

function UpdateMatching(G, M, v)
    Remove vertex v and all incident edges from G to form G'
    if v is not matched in M **then**
        return M
    else
        $u \leftarrow$ the vertex matched with v in M
        $M' \leftarrow M \setminus \{(u, v)\}$
        $P \leftarrow$ augmenting path of (G', M') from u using BFS/DFS
        if P is found **then**
            update M' by flipping the edges along the augmenting path P
        return M'

Correctness and Complexity:

If the deleted vertex v is unmatched, the matching M remains maximum in G'.

If v is matched, removing it leaves vertex u unmatched. According to Berge's theorem, if an augmenting path is found starting from u, flipping the edges restores the matching size; otherwise, M' has already reached the maximum matching size, which means the maximum matching size of G' decreases by one.

The search for an augmenting path runs in $O(n+m)$, which is faster than recomputing the matching from scratch.